

Conformal Cyclic Cosmology Under the Light of TNA

Realizability, Entropy Reset, and Failure of Local Closure in Cyclic Cosmology

Abstract

Roger Penrose's Conformal Cyclic Cosmology (CCC) proposes that the universe does not emerge from an absolute beginning, but instead evolves through an infinite succession of cosmological cycles ("aeons"). In this framework, the asymptotic heat death of one universe becomes conformally equivalent to the Big Bang of the next. By eliminating scale after the evaporation of all black holes, CCC attempts to resolve the low-entropy problem of cosmology without invoking a singular temporal origin.

This paper analyzes CCC through the framework of the Theory of Axiomatic Necessity (TNA). TNA distinguishes between:

- N_0 : the domain of local dynamics, causal propagation, physical evolution, and operational continuity;
- N_1 : the domain of realizability constraints, external selection, boundary conditions, and non-local closure operators.

The paper argues that CCC, despite its mathematical elegance, encounters structural limitations characteristic of Failure of Local Closure. In particular, the conformal transition between aeons functions structurally as a non-derivable reset operator that resembles an implicit N_1 condition embedded within cosmological geometry.

Special attention is given to:

- the entropy reset problem,
- the disappearance of scale,
- the role of conformal transformations,
- the paradox of informational continuity without time,
- and the tension between finite versus infinite aeonic traces in the Cosmic Microwave

The paper proposes that CCC may ultimately reveal not a fully self-contained cosmology, but rather the necessity of a deeper realizability structure beyond purely local physical dynamics.

1. Introduction

Modern cosmology has long struggled with one fundamental question:

Why did the universe begin in a state of extraordinarily low entropy?

Standard cosmological models explain the evolution of entropy after the Big Bang, but not the special initial conditions that made such evolution possible.

Roger Penrose proposed Conformal Cyclic Cosmology (CCC) as an attempt to solve this problem without requiring an absolute beginning of time.

CCC proposes that:

- universes evolve through successive aeons,
- heat death becomes conformally equivalent to a new Big Bang,
- and the disappearance of mass removes all meaningful physical scale.

The result is an eternally recyclable universe.

However, CCC raises deep structural questions:

- What selects the transition between aeons?
- Why does low entropy reappear?
- Can a universe fully generate its own realizability conditions internally?
- Can cyclicity itself provide ontological closure?

These questions align directly with the concerns of TNA.

2. The Core Structure of CCC

CCC proposes the following cosmological sequence:

1. Big Bang and expansion.
2. Formation of stars, galaxies, and black holes.
3. Heat death and extreme dilution.
4. Dominance of black holes.
5. Hawking evaporation of all black holes.
6. Elimination of all massive particles.
7. Loss of physical scale.
8. Conformal geometric rescaling.
9. Emergence of a new aeon.

The critical insight is that:
once all mass disappears, scale loses meaning.

Without mass:

- no clocks exist,
- no rulers exist,
- and no preferred size remains physically definable.

An infinitely large cold universe becomes conformally equivalent to an infinitely small hot universe.

CCC therefore attempts to transform asymptotic infinity into a new cosmological origin.

3. TNA and the Problem of Cosmological Closure

TNA distinguishes between two domains:

3.1 N_0 — Local Physical Dynamics

N_0 includes:

- gravitation,
- thermodynamics,

- quantum processes,
- causal propagation,
- and spacetime evolution.

CCC is fundamentally constructed within N_0 .

Its goal is to explain:

- entropy evolution,
 - cosmological rebirth,
 - and continuity
through internal physical processes alone.
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3.2 N_1 — Realizability Constraints

N_1 represents:

- boundary conditions,
- selection operators,
- realizability constraints,
- and non-local closure structures.

N_1 becomes necessary whenever:

local dynamics cannot fully determine their own realizability.

From the TNA perspective, CCC attempts to reduce cosmological closure entirely to N_0 .

This creates structural tension.

4. The Entropy Reset Problem

The central achievement claimed by CCC is the restoration of low entropy.

Penrose argues that:

- gravitational degrees of freedom contain most entropy,
- and once all mass disappears,

- gravitational entropy effectively vanishes.

The universe becomes smooth again.

However, from a TNA perspective, a critical question emerges:

Why should multiplicity collapse into smoothness?

A highly evolved universe should contain:

- enormous informational complexity,
- chaotic multiplicity,
- and accumulated structural asymmetry.

Yet CCC requires:

a reset toward highly ordered realizability.

This suggests the presence of a structural selector.

Under TNA, the conformal reset behaves analogously to an implicit N_1 operator:
a condition enabling renewed realizability despite maximal thermodynamic evolution.

5. Conformal Degeneracy and Loss of Scale

The most radical aspect of CCC is the disappearance of scale.

Once all mass vanishes:

- spacetime loses measurable differentiation,
- size becomes undefined,
- and temporal direction becomes physically meaningless.

From TNA, this implies something profound:

N_0 loses the ability to internally self-index its own evolution.

The system can no longer distinguish:

- large from small,
- past from future,

- expansion from contraction.

The universe approaches a state of conformal degeneracy.

This resembles a breakdown of local closure.

The transition between aeons therefore acts not merely as continuity, but as:

- realizability re-selection,
 - structural renormalization,
 - and ontological reset.
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6. Informational Continuity Without Time

CCC proposes that traces of previous aeons may remain visible within the Cosmic Microwave Background.

This creates a paradox.

The theory simultaneously requires:

- the disappearance of conventional temporality, while also preserving:
- informational continuity across aeons.

Thus CCC implicitly requires:
memory without classical time.

This exceeds standard local physical propagation.

From a TNA perspective, this suggests:

- a deeper continuity structure,
- potentially operating beyond ordinary spacetime locality.

In other words:
CCC may require a non-local realizability substrate.

7. The Problem of Aeonic Traces

Penrose argues that previous aeons may leave observable “scars” in the CMB: concentric low-variance circles or anomalous correlations.

This creates a structural dilemma.

7.1 Finite Aeons

If only finitely many aeons exist:

- then a first aeon must exist,
- and CCC fails to eliminate the origin problem.

A first low-entropy state still requires explanation.

7.2 Infinite Aeons

If infinitely many aeons exist:

- one might expect infinitely layered cosmological residues,
- far greater statistical saturation,
- or overwhelming trace density within the CMB.

However:

observations do not clearly support infinite residue accumulation.

This tension suggests that:

CCC struggles to reconcile infinite cyclicity with finite observable structure.

From TNA, this resembles a classical symptom of Failure of Local Closure: the system attempts self-sufficiency while remaining unable to fully stabilize its own realizability conditions.

8. Penrose and the Limits of Reductionism

An intriguing philosophical parallel emerges.

In works such as:

- The Emperor's New Mind
- Shadows of the Mind

Penrose argues that:

human consciousness cannot be fully reduced to computation.

Yet CCC appears to encounter a similar limitation cosmologically.

The universe itself may not be fully self-derivable.

Thus:

Penrose resists computational closure in cognition,

while CCC indirectly encounters cosmological non-closure.

This convergence is highly significant from a TNA perspective.

9. CCC as Implicit N_1 Cosmology

From the standpoint of TNA, CCC can be reinterpreted structurally as:

- a universe attempting self-realization through cyclic dynamics, while simultaneously requiring:
- a hidden reset operator,
- non-local conformal equivalence,
- and repeated low-entropy relicensing.

The conformal transition functions structurally like:

- a selector,
- a realizability filter,
- and a boundary-induced renormalization mechanism.

In TNA terms:

CCC may unknowingly encode N_1 within geometry itself.

The theory therefore does not eliminate transcendence or boundary conditions.
Instead:
it geometrizes them.

10. Conclusion

Conformal Cyclic Cosmology is among the most mathematically elegant attempts to explain cosmological continuity without invoking an absolute beginning.

However, under the framework of TNA, CCC reveals deep structural tensions.

The theory successfully:

- removes classical temporal origins,
- eliminates absolute scale,
- and cyclically reconnects cosmological evolution.

Yet it continues to require:

- entropy reset,
- conformal re-selection,
- informational continuity without time,
- and renewed realizability after thermodynamic exhaustion.

These features suggest that:

local physical dynamics alone may remain insufficient to fully ground cosmological closure.

CCC therefore appears not as a fully self-contained cosmology,
but as a sophisticated model approaching the limits of N_0 self-derivability.

Under TNA, the conformal transition between aeons behaves structurally like an implicit N_1 operator:

a deeper condition enabling the universe to remain realizable across cycles.

The universe may thus be:

- locally physical,
- globally open,
- and continuously dependent upon a deeper structure of realizability beyond purely internal dynamics.